
The central limit theorem

Antonio de Leon

Probability and Statistics for Engineers Fall 2026

Monday, November 16

Midterm 2

Wednesday, November 18

Sums, averages, and standardization

Friday, November 20

Approximation and interpretation

Course support and deadlines

Ed Discussion: questions, announcements, and private messages. **No Canvas Inbox.**

[Open Ed from Canvas](#) All times Pacific

[Full contact details and meeting links](#)

Instructor / TAs	Office hours	Optional, ungraded discussions
Antonio de Leon	Fri 3–4 p.m. (in person) GSC Room 204	—
Jeannine Romero	Tue / Thu 10–11 a.m.	01A: Fri 9:20–10:25 a.m. R Carson Acad 240 (in person)
Ehsan Mirafzali	Mon 3–5 p.m.	01B: Fri 10:40–11:45 a.m. (Zoom)
Seokjun Choi	Wed 3–4 p.m. 5:05–6:05 p.m.	01C: Wed 4–5:05 p.m. (Zoom)

Upcoming deadlines

HW and quizzes: **10 p.m. Pacific**

Midterm 2	Mon, Nov 16	During class; Classroom Unit 002
Quiz 8	Sun, Nov 22	Canvas
HW 6	Tue, Nov 24	Gradescope

GSC: Graduate Student Commons. TA office-hour times [link to Zoom](#).

HW: 24-hour grace period. Quizzes: no grace period. Approved accommodations apply.

Did You Know?

Fresh

Dried

JALAPEÑO



CHIPOTLE



POBLANO



ANCHO



CHILACA



PASILLA



MIRASOL



COLORADO



SERRANO



GUAJILLO



SERRANO



CHILE SECO



BOLA



CASCABEL



Midterm 2

Monday, November 16

55 minutes of examination work. Use the separate examination and formula sheet.

Scope	Required material taught through November 9, with earlier foundations.
Core	Expectation, variance, joint and conditional laws, covariance, elementary CDF transformations, Markov and Chebyshev bounds, and the finite-variance weak LLN.
Excluded	CLT, inference, and stochastic processes.

No new lecture material in this meeting.

A sum and an average

Let $X_1, \dots, X_n$ be iid with mean μ and variance σ^2 .

$$S_n = X_1 + \dots + X_n, \quad \bar{X}_n = \frac{S_n}{n}.$$

If each observation measures time in seconds, what does each quantity measure?

Predict the scales first. Then derive them.

Mean and variance of a sum

Assume $\mathbb{E}[X_i] = \mu$ and $\text{Var}(X_i) = \sigma^2 < \infty$ for iid observations. Derive the moments of S_n and $\bar{X}_n$. Identify where independence enters.

The variance of an average contains a factor $1/n^2$ before simplification.

One standardized quantity

Suppose $0 < \sigma^2 < \infty$. Center and divide by the standard deviation:

$$Z_n = \frac{S_n - n\mu}{\sigma\sqrt{n}}.$$

Express Z_n using $\bar{X}_n$, then derive its mean and variance.

The central limit theorem

Let $X_1, X_2, \dots$ be iid, with

$$\mathbb{E}[X_i] = \mu, \quad 0 < \sigma^2 = \text{Var}(X_i) < \infty.$$

For every fixed real z ,

$$\lim_{n \rightarrow \infty} P\left(\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \leq z\right) = \Phi(z).$$

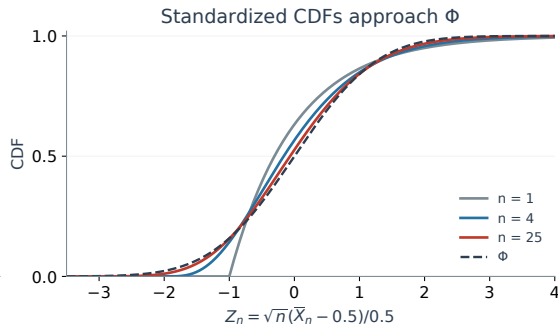
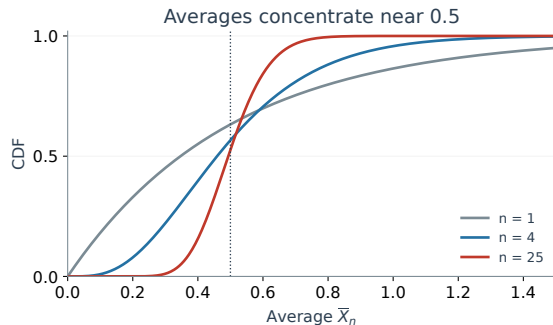
Here $\Phi(z) = P(G \leq z)$ for $G \sim N(0, 1)$.

The standardized CDF approaches the standard normal CDF.

Theorem stated without proof. Harchol-Balter, §9.4; DeGroot–Schervish, §6.3.

Concentration and standardized fluctuations

X_i are iid exponential with rate 2, so $\mu = \sigma = 1/2$.



LLN: $\bar{X}_n$ concentrates near μ . CLT: the CDF of Z_n approaches Φ .

Exponential measurements

Model independent component times by $X_i \stackrel{\text{iid}}{\sim} \text{Exp}(2)$, with rate 2 per second. Recall

$$f_X(x) = 2e^{-2x} I_{\{x>0\}}, \quad \mu = \frac{1}{2}, \quad \sigma^2 = \frac{1}{4}.$$

For $n = 100$, approximate

$$P(0.45 < \bar{X}_n < 0.55).$$

Probability for an average

$X_i \stackrel{\text{iid}}{\sim} \text{Exp}(2)$, $n = 100$, $\mu = \sigma = 1/2$. Find the sampling SD and translate $0.45 < \bar{X}_n < 0.55$ to the Z_n scale.

$\Phi(1) \approx 0.841345$. The normal replacement is the only approximate step.

The same event for the sum

With $n = 100$, $S_{100} = 100\bar{X}_{100}$.

$$\{0.45 < \bar{X}_{100} < 0.55\} = \{45 < S_{100} < 55\}.$$

Quantity	Mean	Standard deviation
$\bar{X}_{100}$	0.5	0.05
S_{100}	50	5

Check the standardized endpoints using the sum.

Four times as many observations

Keep $X_i \stackrel{\text{iid}}{\sim} \text{Exp}(2)$ and the same tolerance. For $n = 400$, approximate $P(0.45 < \bar{X}_n < 0.55)$. Explain the change from $n = 100$ before calculating.

$$\Phi(2) \approx 0.977250.$$

Equality, bound, limit, approximation

Assume iid observations with mean μ and $0 < \sigma^2 < \infty$. Identify the status of each statement.

Statement	Status
$\text{Var}(\bar{X}_n) = \sigma^2/n$	
$P(\bar{X}_n - \mu \geq \varepsilon) \leq \sigma^2/(n\varepsilon^2)$	
$P(Z_n \leq z) \rightarrow \Phi(z)$ for fixed z	
$P(Z_{100} \leq 1) \approx \Phi(1)$	

Center and scale

iid observations have mean μ and variance $\sigma^2 > 0$. Complete the denominators:

$$\frac{S_n - n\mu}{\quad}, \quad \frac{\bar{X}_n - \mu}{\quad}.$$

Variance of a Bernoulli variable

Let I equal 1 with probability p and 0 with probability $1 - p$. Use $I^2 = I$ to derive its variance.

An indicator counts one success. Its square has the same value.

Mean and variance of a count

Let $I_1, \dots, I_n$ be iid Bernoulli(p), with $0 < p < 1$, and

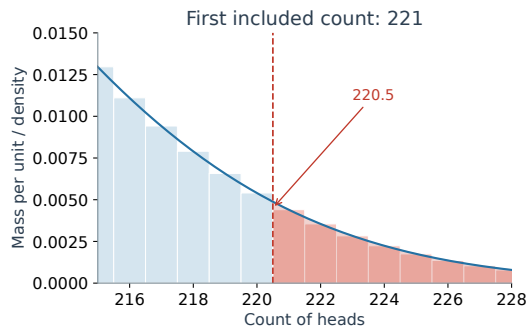
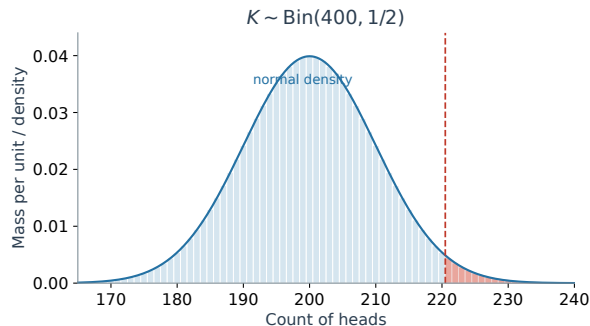
$$K_n = \sum_{i=1}^n I_i \sim \text{Bin}(n, p).$$

Derive the moments and the standardized count.

Check the trial assumptions before using a binomial model.

Counts occupy cells

For integer k , $\{K > k\} = \{K \geq k + 1\}$. The first included cell starts at $k + 1/2$.



$$\begin{array}{ccc} K \leq k & K \geq k & a \leq K \leq b \\ \text{up to } k + 1/2 & \text{from } k - 1/2 & \text{from } a - 1/2 \text{ to } b + 1/2 \end{array}$$

Each unit-width bar has area equal to its probability mass.

More than 220 heads

A fair coin is tossed independently 400 times. Let K count the heads.

$$K \sim \text{Bin}(400, 1/2).$$

Approximate $P(K > 220)$.

Adapted from the supplied STAT131 Discussion 10.

Normal approximation to the tail

$K \sim \text{Bin}(400, 1/2)$ has mean 200 and SD 10. Use the boundary 220.5 for $\{K > 220\}$.

Method	Probability	Absolute error
Exact sum (rounded)	0.0201154	—
Normal, no correction	0.0227501	0.0026348
Normal, corrected	0.0201822	0.0000668

Exact benchmark: $2^{-400} \sum_{j=221}^{400} \binom{400}{j} \cdot \Phi(2.05) \approx 0.979818$.

A large sample can still give a poor approximation

Let $K \sim \text{Bin}(1000, 0.001)$. Then $\mathbb{E}[K] = 1$ and $\text{Var}(K) = 0.999$.

$$P(K \geq 1) = 1 - (0.999)^{1000} \approx 0.632305.$$

Corrected normal approximation: $1 - \Phi\left(\frac{0.5 - 1}{\sqrt{0.999}}\right) \approx 0.691551.$

The sample size alone cannot certify accuracy.

Planning a sample size

lid rod lengths have mean 50 mm and known SD 8 mm. Choose n for a target probability 0.95 of $|\bar{X}_n - 50| < 2$ mm.

62 is an approximate planning choice. 320 is sufficient from the variance alone.

A count threshold

Independent tests each pass with probability $1/2$. Among 100 tests, let K be the number that pass. Approximate $P(K \geq 60)$. State the model, identify the first included cell boundary, and interpret your answer.

$$\Phi(1.9) \approx 0.971283.$$

What the approximation establishes

Assumptions	iid observations with positive finite variance.
Quantity	Centered sum or average, divided by its own SD.
Finite sample	A normal probability approximation whose accuracy needs judgment.
Sampling SD	$\sigma/\sqrt{n}$ for $\bar{X}_n$, often called its standard error.

Next lecture: use sampling variability to construct an interval for a mean.

Optional: independent signal noise

$X_i \stackrel{\text{iid}}{\sim} \text{Unif}(-1, 1)$ models additive component noise. For $n = 100$, approximate $P(|S_n| < 10)$ and compare with Chebyshev.

Harchol-Balter, Example 9.9, pp.177–178. $\Phi(\sqrt{3}) \approx 0.958368$.

Optional: a smaller tail

Let $K \sim \text{Bin}(900, 1/2)$. Approximate $P(K > 495)$. Compare the direct normal calculation with the continuity-corrected result.

DeGroot–Schervish, Example 6.3.2 and §6.4. Supplied Lec10, frames 21–22.

Optional: one count value

For $K \sim \text{Bin}(20, 1/2)$, approximate $P(K = 10)$. Which interval represents the mass at ten?

DeGroot–Schervish, Example 6.4.4.

Optional: repeated measurements

Let X_i be iid unbiased measurements of a true value θ , with SD $\sigma > 0$. For $n = 25$, estimate $P(|\bar{X}_n - \theta| < \sigma/4)$. Give a Chebyshev lower bound as well.

Supplied STAT131 problems, §4.2, Q1. $\Phi(1.25) \approx 0.894350$.

Optional: a 99 percent precision target

Keep the same measurement model and tolerance $\sigma/4$. Find a normal planning size and a Chebyshev sufficient size for a target probability 0.99.

Supplied STAT131 problems, §4.2, Q2.

Selected reading

CLT and standardization

Harchol-Balter, *Introduction to Probability for Computing*, supplied 2024 version, §9.4, pp.176–178.

DeGroot–Schervish, *Probability and Statistics*, 4th ed., §6.3, pp.361–363.

Continuity correction

DeGroot–Schervish, §6.4, pp.371–374.

Optional practice

Week 8 optional practice and checked answers. The notes contain the required statements and derivations.

The CLT proof, delta method, and more general limit theorems are outside this week's required material.